AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 34 · STUDENT LESSON

Test Edge Cases and Fairness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Week 16: review can be an expected result rather than a mistake.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Choose tests just below, at and above a decision boundary. Try the recall before opening notes.

Your learning target

I can choose tests just below, at and above a decision **boundary**.

I can distinguish valid no-match requests from **invalid input**.

I can use labelled catalogue variations to examine previously untested paths.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

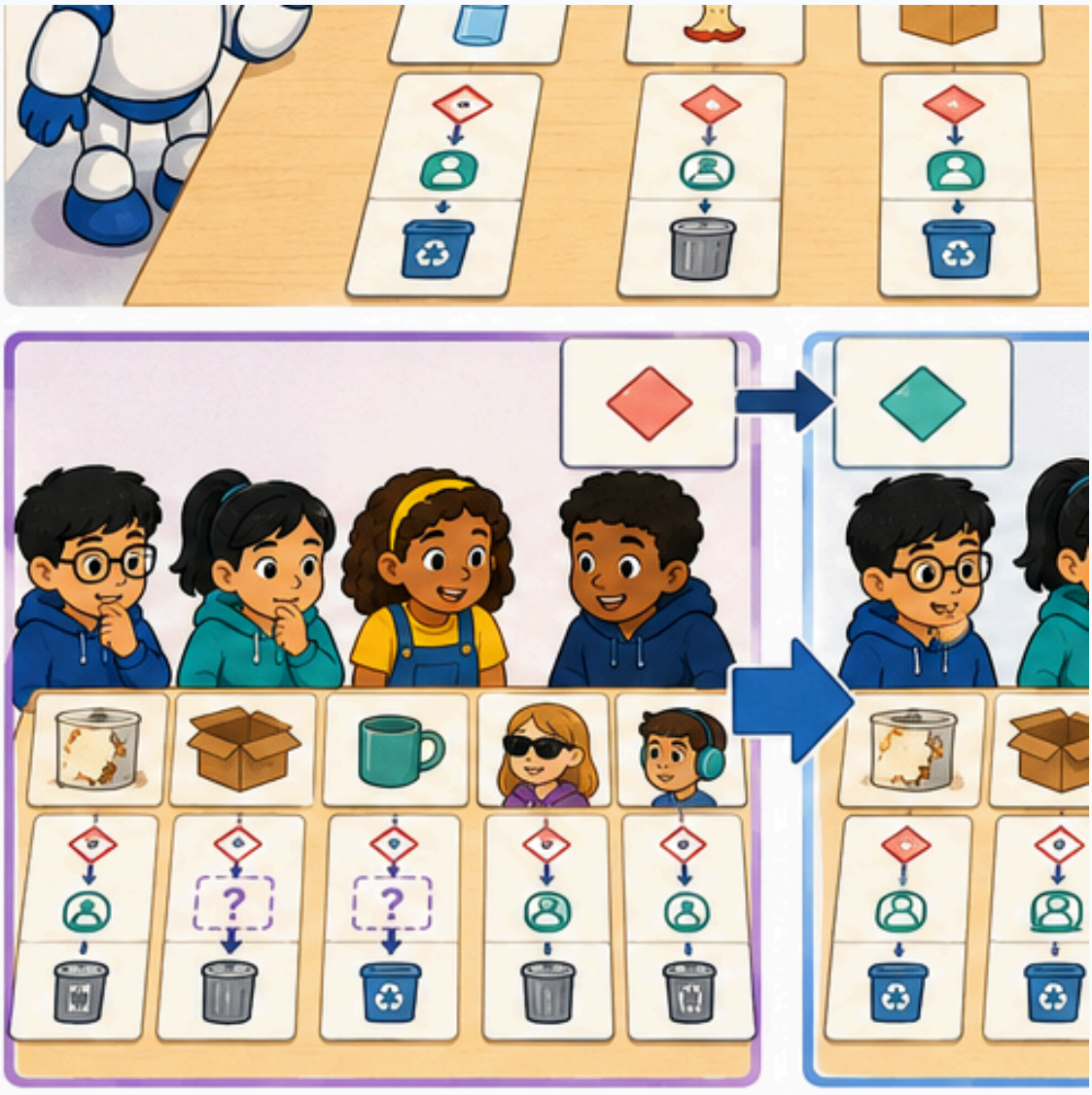
Without opening the old lesson, explain one key idea from Test Normal Cases. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Design the Decision **Process**. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Unusual and **underrepresented** cases reveal **hidden assumption**s.



Your target

LEARNING TARGET Before testing, define outcomes for clean paper, metal foil, mixed packaging and an unreadable card. The team later changes mixed packaging's expected result to match its sorter.

Explain the evaluation problem.

Learner name:

Readiness for the connected explanation

Why can four successful routine cases leave a mistake undiscovered?

This week's end goal

By the end, I can design **boundary** and coverage tests instead of assuming routine success is enough.

STUDENT LESSON · 2 OF 18

Notice the evidence

Investigate the supplied Test Edge Cases and Fairness case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY A worn notebook, multilingual label, partly hidden item, and object **fitting** two bins reveal assumptions normal tests miss.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Routine examples can miss a mistake at an exact limit. This week you will choose revealing cases around those limits and vary the paper catalogue to reach later successful **output**s.

Write expected results from the requirements before operating the prototype. Keep its saved instructions and each test record so a failure remains useful **evidence**.

Catalogue B version 1 is unchanged: B1 Bookmark drawing, 10 minutes, quiet Yes; B2 Story circle, 8 minutes, quiet No; B3 Pattern puzzle, **12 minutes**, quiet Yes; B4 Tower challenge, 15 minutes, quiet No. Search in that order.

These fictional paper values assume materials are available. **Input**s are whole minutes from 0 to 30 **inclusive** and Quiet required Yes or No. No allows quiet or nonquiet activities.

Time fits when required minutes are no more than available minutes, including equality. The quiet condition must also hold. Select the first **fitting** row with a correct reason, or report no match after every row fails.

Invalid or missing **input** requires clarification before searching. P1 validates, resets to the first row, checks both conditions, advances after failure and stops at the first fit or catalogue exhaustion.

No **output** books or publishes anything.

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Test Edge Cases and Fairness. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.



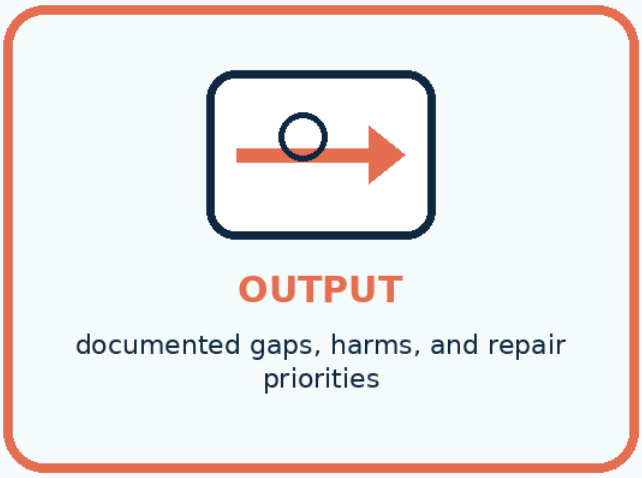
INPUT

planned edge and subgroup cases



PROCESS

run same version and compare error patterns



INPUT planned edge and subgroup cases

PROCESS run same version and compare error patterns

OUTPUT documented gaps, harms, and repair priorities

Key idea

CORE IDEA Unusual and **underrepresented** cases reveal **hidden assumption**s.

How the mechanism works

UNDER THE HOOD Edge testing explores boundaries, rare combinations, missing values, adversarial ambiguity, and subgroup conditions. Fairness analysis asks whether error burdens differ and whether fallback paths remain usable.

Draw a fresh example with all three stages:

A closer explanation

Three requests around ten minutes

AVAILABLE TIME	CHECK B1 TIME	RESULT FOR QUIET YES
9 minutes	10 is too long	No match in catalogue B
10 minutes	10 fits exactly	Suggest B1
11 minutes	10 fits with time left	Suggest B1

Keep Quiet required Yes and catalogue B version 1 fixed while changing available time.

For quiet requests, 10 minutes is the first time a catalogue activity fits. Test 9, 10 and 11 with Yes so only the time changes.

A faulty comparison S version 0 uses strictly less than instead of no more than for time, while keeping the other P1 rules. It rejects B1 at exactly 10.

With quiet required, every later row also fails, so it incorrectly reports no match.

S can pass all four routine requests from Week 33: 11 Yes, 9 No, 6 No and 9 Yes. None has a **fitting** activity whose duration equals the request. The 10 Yes case reveals a gap those passes did not close.

Another useful **boundary** is 8 minutes with No: equality should allow B2, while 7 No has no match. Keep both quiet values in your wider plan because No does not mean nonquiet only.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

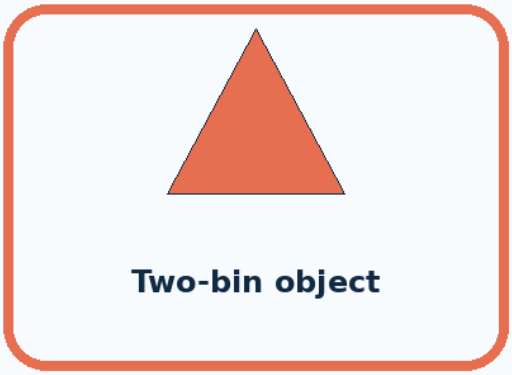
> Compare the supplied cases for Test Edge Cases and Fairness. Identify the detail that changes the conclusion.

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — Worn label Clue: text partly missing. Meaning: degraded **input**.

The case to investigate

CASE 2 — Two-bin object Clue: categories overlap. Meaning: **boundary** case.

The case to investigate

CASE 3 — Different language Clue: label format changes. Meaning: access/fairness case.

The case to investigate

CASE 4 — Blank card Clue: required **input** absent. Meaning: missing-data case.

Keep this boundary clear

BOUNDARY Edge cases are not excuses to blame users for being unusual; they are **evidence** about design limits.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: Faulty code uses duration strictly below available time.

Predicted result

Prediction: Request 6 now wrongly returns none.

Why the result follows

Reason: A's duration equals the available time and should be eligible under at most.

Words that unlock the idea

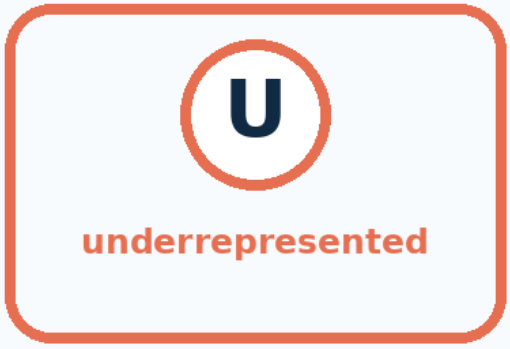
> Use the vocabulary of Test Edge Cases and Fairness to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
edge case	unusual input near a system limit	
underrepresented	rare or absent in design evidence	
subgroup test	check for one relevant condition or user group	
hidden assumption	unstated belief revealed by failure	

Misconception detective

Find the unsupported claim

CLAIM TO REPAIR Rare cases do not matter.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Boundary	A value where an input changes category or a condition changes truth
Inclusive	Including the stated endpoint or equal value
Invalid input	A value that does not meet the agreed input rules
Test variation	A documented change in the conditions used for a test
Coverage gap	A relevant condition or behaviour the current tests have not examined

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STUDENT LESSON · 6 OF 18

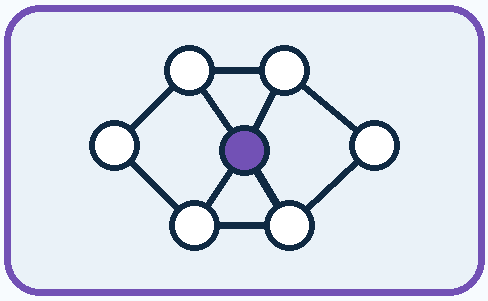
Follow a worked example

> Follow the worked trace for Test Edge Cases and Fairness, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE A worn notebook, multilingual label, partly hidden item, and object **fitting** two bins reveal assumptions normal tests miss.

Step 1

STEP 1 — OBSERVE planned edge and subgroup cases

Step 2

STEP 2 — TRACE run same version and compare error patterns

Step 3

STEP 3 — CONCLUDE documented gaps, harms, and repair priorities

Model answer

MODEL REASONING Use the same frozen prototype, log outcomes by condition, and compare error types. Prioritize failures with high burden or no usable fallback, then design a targeted repair without stereotyping users.

A claim to avoid

DO NOT OVERCLAIM **Edge case**s are not excuses to blame users for being unusual; they are **evidence** about design limits.

What new **evidence** could change this answer?

A closer explanation

Request using catalogue B	Expected response	Reason
0 minutes Yes	No match	Zero is valid; every activity needs more time
30 minutes Yes	B1 with a matching reason	30 is valid; B1 is the first fit
31 minutes Yes	Clarify minutes; no search	31 is outside the accepted range
Blank minutes with Yes	Clarify minutes; no search	The time field is missing
10.5 minutes Yes	Clarify minutes; no search	The accepted time must be a whole number
10 minutes with blank quiet	Clarify quiet; no search	Choose Yes or No before searching

Worked explanation · Test Edge Cases and Fairness

The accepted range and activity durations are different kinds of **boundary**. Zero belongs to the valid range even though no activity fits. Thirty also belongs to it.

A negative time, such as minus one minute, is below zero and invalid. Do not change a blank to zero, round 10.5 to 10 or turn a blank quiet choice into No. Ask for the field to be corrected and start a new validated run.

Worked explanation · Test Edge Cases and Fairness

A test record should say whether a search is expected to begin. Clarification should name the field needing correction; it should not claim the catalogue has no suitable activity.

Test an unrecognised quiet value such as Maybe as well as a missing value. These check different forms of unusable **input**.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Test Edge Cases and Fairness

A fictional craft-book chooser accepts requests for whole available minutes from 0 to 20.

The facts we will use

Supplied facts

Booklet A takes 6 minutes; B takes 12. Return first duration at most available time.

Supplied facts

Check 5,6,7; also 0,20,21 and blank. Same catalogue applies to every user.

AI ACADEMY / AI-1/34

Test Edge Cases and Fairness

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Near a boundary

At 5: none. At 6 and 7: A.



2. Valid extremes

0: none. 20: A, because first-fit order remains fixed.



3. Invalid inputs

21 and blank require clarification, not ordinary search.



4. Coverage record

Record each user condition tested; do not invent unrun outcomes.

CHANGE ONE THING

Faulty code uses duration strictly below available time.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Near a boundary	At 5: none. At 6 and 7: A.
2. Valid extremes	0: none. 20: A, because first-fit order remains fixed.
3. Invalid inputs	21 and blank require clarification, not ordinary search.
4. Coverage record	Record each user condition tested; do not invent unrun outcomes.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Adjacent values below, at and above 6 expose an accidental strict inequality. Zero is valid even though no activity fits.

The top valid **input** does not require selecting the longest activity: catalogue order still controls selection. Coverage also asks whose conditions were tested, which cannot be inferred from these numbers alone.

What this example does not establish

Keep the stated limit

Equal **written rule**s do not prove equally good outcomes for every group; actual condition-specific **evidence** is still needed.

Keep the stated limit

Read the labelled picture: Which single **boundary** run most directly detects the strict-comparison bug, and why?

Keep the stated limit

Change one thing: Faulty code uses duration strictly below available time. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Test Edge Cases and Fairness claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept **boundary**.

Case	Evidence	What it shows
Worn label	text partly missing	degraded input
Two-bin object	categories overlap	boundary case
Different language	label format changes	access/fairness case
Blank card	required input absent	missing-data case

Repair two tempting claims

Challenge claim A

CLAIM A Rare cases do not matter.

Repair:

Challenge claim B

CLAIM B One special test represents every member of a subgroup.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

Label each case by its purpose: time equality, accepted-range endpoint, invalid field or catalogue variation. Record the exact catalogue version and request, the expected **output** and explanation, and what actually happened.

Compare the prototype with the manual baseline under identical conditions. Count every attempted case, retain failures, and distinguish cases not yet run from successful tests.

Check the claim against the facts

These are development tests that can guide repairs. Keep any final evaluation cases reserved by your teacher separate until the prototype version is fixed for evaluation.

Even a broad development set does not establish real-world enjoyment, material availability or speed. Next week you will repair a located defect and check whether the change breaks behaviour that worked before.

S passes the four routine cases but fails 10 Yes by reporting no match. Explain which comparison to inspect and why changing the expected answer to no match would be wrong. Give one additional equality case and one invalid-input case, including their expected outcomes.

Practise together

> Practise Test Edge Cases and Fairness with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Unusual and **underrepresented** cases reveal **hidden assumption**s.

Use a fresh example to explain **edge case**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: One special test represents every member of a subgroup.

Invent one **edge case** and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

In B version 1, earlier B1 always fits whenever B3 or B4 could be chosen. To test a successful later-row path, make a labelled copy of the catalogue.

Keep the **input** rules, both compatibility conditions and first-match rule unchanged. Both the prototype and manual baseline must read the same test copy.

Do not overwrite the original catalogue or reuse its expected answers for the changed data.

Test copy B version 2 changes only B1's duration from 10 to 20 minutes. All other rows, quiet labels and order stay as in version 1. For 12 Yes, B1 is now too long, B2 is not quiet, and B3 fits exactly and is quiet.

The expected first suggestion is B3. This checks a third-row success and equality under a documented variation.

> **Make test copy B version 3 with durations B1 20, B2 18, B3 17 and B4 15 minutes. Keep the original quiet labels and order. For 15 No, trace the rows and give the required output and reason. Then use 15 Yes with the same copy. Explain why the outcomes differ and what must be written on both test records.**

Practise with less help: Check the validation boundary

Use workbook W2 for the connected example "Testing boundaries and varied conditions". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Testing boundaries and varied conditions. Proposed explanation: “At the limit means outside the rule”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Which single **boundary** run most directly detects the strict-comparison bug, and why?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

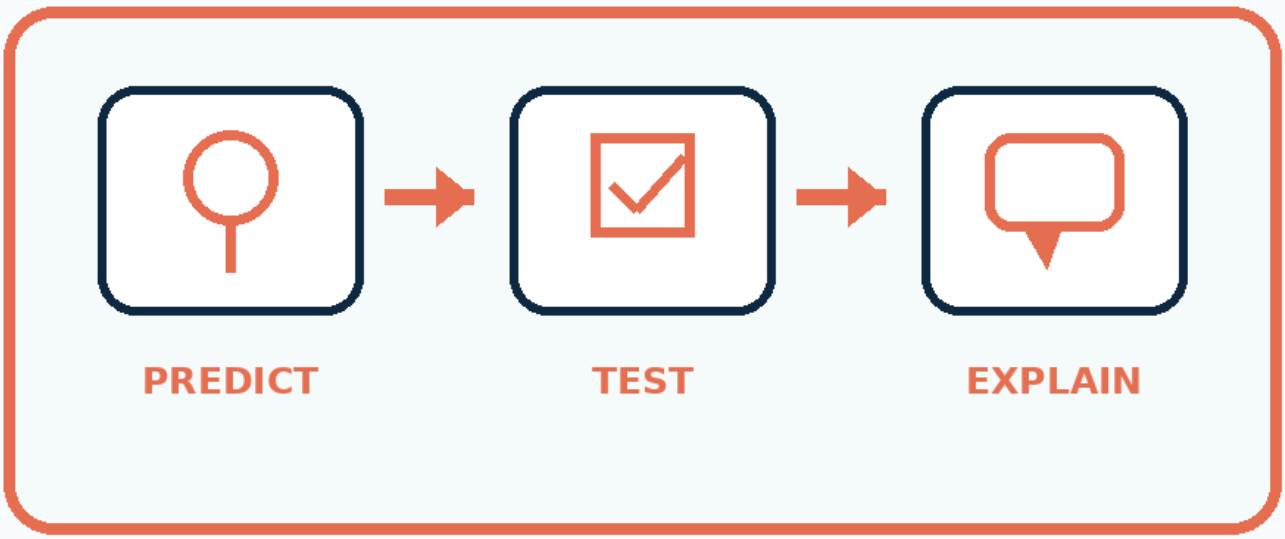
Run the investigation

> Investigate Test Edge Cases and Fairness. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Edge and Fairness Test Pack

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Add missing, ambiguous, and **subgroup test** cards.

3. Compare cases

Keep a normal case and an **edge case** clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

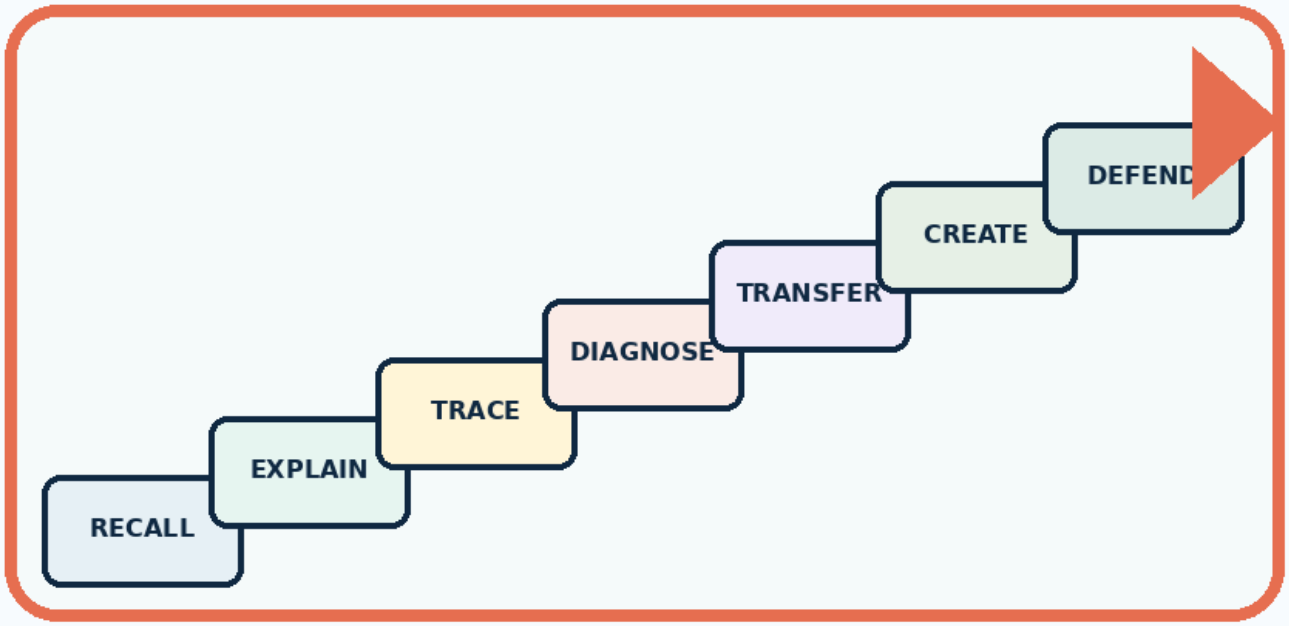
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Test Edge Cases and Fairness. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **edge case**.

EXPLAIN Unusual and **underrepresented** cases reveal **hidden assumption**s.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: Rare cases do not matter.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

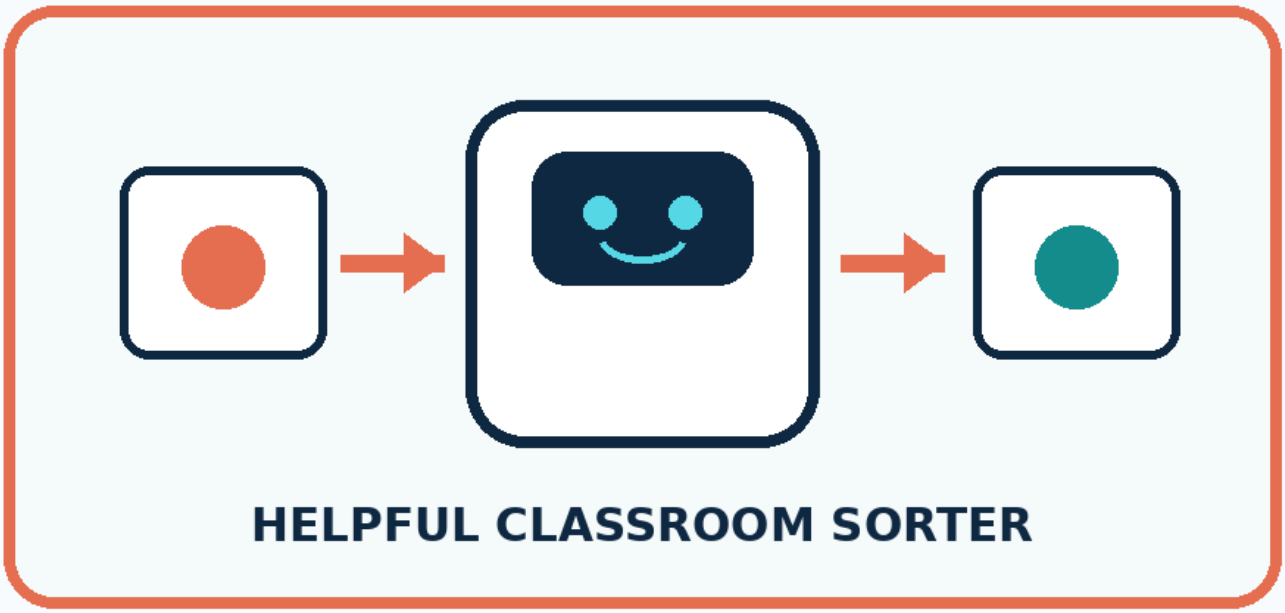
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



THIS WEEK'S CHECKPOINT Expand the sorter test suite with edge, missing-data, and fairness cases plus expected safe behavior.

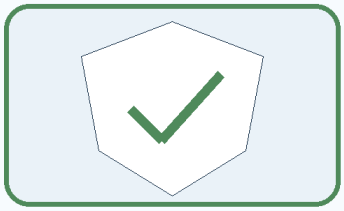
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and **edge case**s will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN **Edge case**s are not excuses to blame users for being unusual; they are **evidence** about design limits.

My project decision:

Evidence for your project

Add sorter **boundary** cases, missing fields and relevant user conditions to the test plan before making fairness claims.

STUDENT LESSON · 12 OF 18

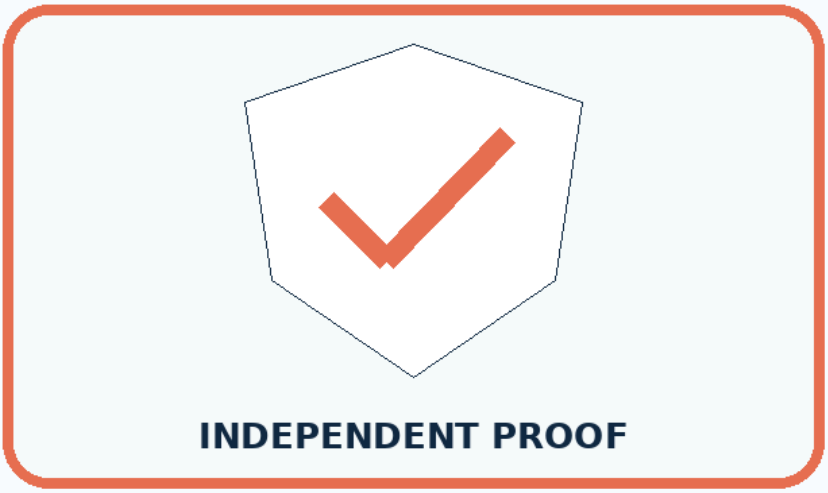
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Add missing, ambiguous, and **subgroup test** cards, compare results, and name one **hidden assumption**.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK Use the same frozen prototype, log outcomes by condition, and compare error types. Prioritize failures with high burden or no usable fallback, then design a targeted repair without stereotyping users.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Improve One Cause.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Test Edge Cases and Fairness case for supported practice. Repeated examples do not become fresh assessments.

Week 34 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A sorter handles clean paper but fails on crumpled paper and mixed packaging.

The question

Design revealing tests.

Worked explanation

Use separately labelled clean, crumpled and mixed cases with expected **output**s or review actions declared first. Compare conditions.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Test conditions that reveal limitations

Evidence or stage	Value or result
Clean paper	Routine case
Crumpled paper	Changed condition
Mixed packaging	Ambiguous material
Expected action	Declare tray or review before testing

A sorter handles clean paper but fails on crumpled paper and mixed packaging.

Design revealing tests.

Use separately labelled clean, crumpled and mixed cases with expected **output**s or review actions declared first. Compare conditions.

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Find an independent capacity boundary

A new equipment selector lists E1 first, capacity 4; E2 second, capacity 8. Valid group size is a whole number from 1 to 10.

Choose the first item whose capacity is at least the group size, or no match; **invalid input** needs clarification. A faulty version uses capacity strictly greater than group size.

Independently give correct and faulty **output**s for groups 3, 4 and 5, then for group 8. Explain the defect and give the correct response to group 0. No booking occurs.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 34 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A sorter works on clear photos. Test a blurred pencil, a dim book and an unfamiliar lunch box. What makes these useful tests?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a **boundary** or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Expand the sorter test suite with edge, missing-data, and fairness cases plus expected safe behavior.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Before testing, define outcomes for clean paper, metal foil, mixed packaging and an unreadable card. The team later changes mixed packaging's expected result to match its sorter. Explain the evaluation problem.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Test Edge Cases and Fairness, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional craft-book chooser accepts requests for whole available minutes from 0 to 20.

By the end, I can	How I will show it
Choose tests just below, at and above a decision boundary	Explain the deciding step, show a trace or test, and state one limit.
Distinguish valid no-match requests from invalid input	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Use labelled catalogue variations to examine previously untested paths	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Test Edge Cases and Fairness without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Choose tests just below, at and above a decision **boundary**

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional craft-book chooser accepts requests for whole available minutes from 0 to 20.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Record each user condition tested; do not invent unrun outcomes.

1. Near a **boundary**

2. **Invalid input**s

3. Coverage record

4. Valid extremes

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Coverage record Record each user condition tested; do not invent unrun outcomes. Adjacent values below, at and above 6 expose an accidental strict inequality.

Zero is valid even though no activity fits. The top valid **input** does not require selecting the longest activity: catalogue order still controls selection.

Coverage also asks whose conditions were tested, which cannot be inferred from these numbers alone.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Faulty code uses duration strictly below available time. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Request 6 now wrongly returns none.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Request 6 now wrongly returns none. A's duration equals the available time and should be eligible under at most.

Repair a missing building block

Review Test Normal Cases (ai-1/33). Goal: Plan routine cases with expected results derived from the requirements. Short practice: Why is none correct for 8 but incorrect for 6? Use both capacity cards..

Return to this week and explain what you changed.

Review Design the Decision **Process** (ai-1/31). Goal: Design a decision **process** that covers valid, invalid and no-match requests.

Short practice: Where does the first original rejection occur, and why is K3 not originally selected?. Return to this week and explain what you changed.

Review Humans Set Boundaries (ai-1/28). Goal: Distinguish a system's capability, the correctness of its **output** and permission to take an action. Short practice: Which original proposal is authorised?

Why do the other two need separate approval?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

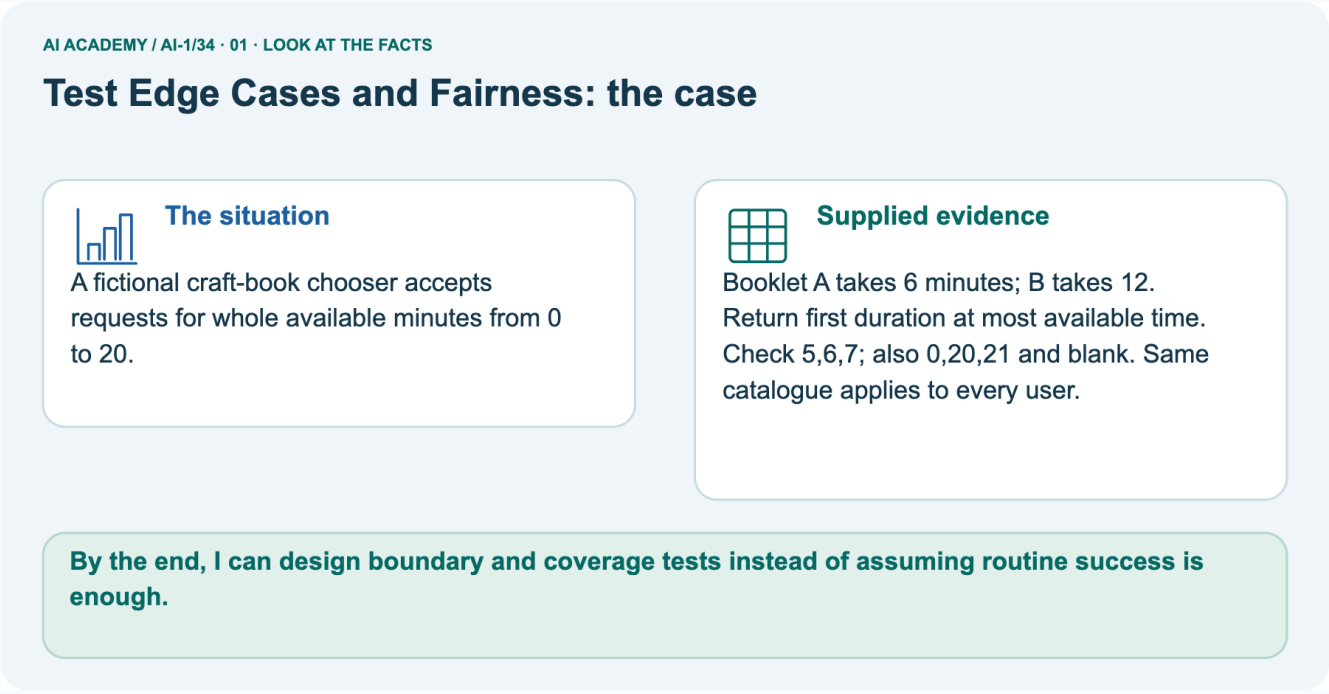
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Test Edge Cases and Fairness graphic using its labels, deciding evidence and relevant limitation.

01. Test Edge Cases and Fairness: the case



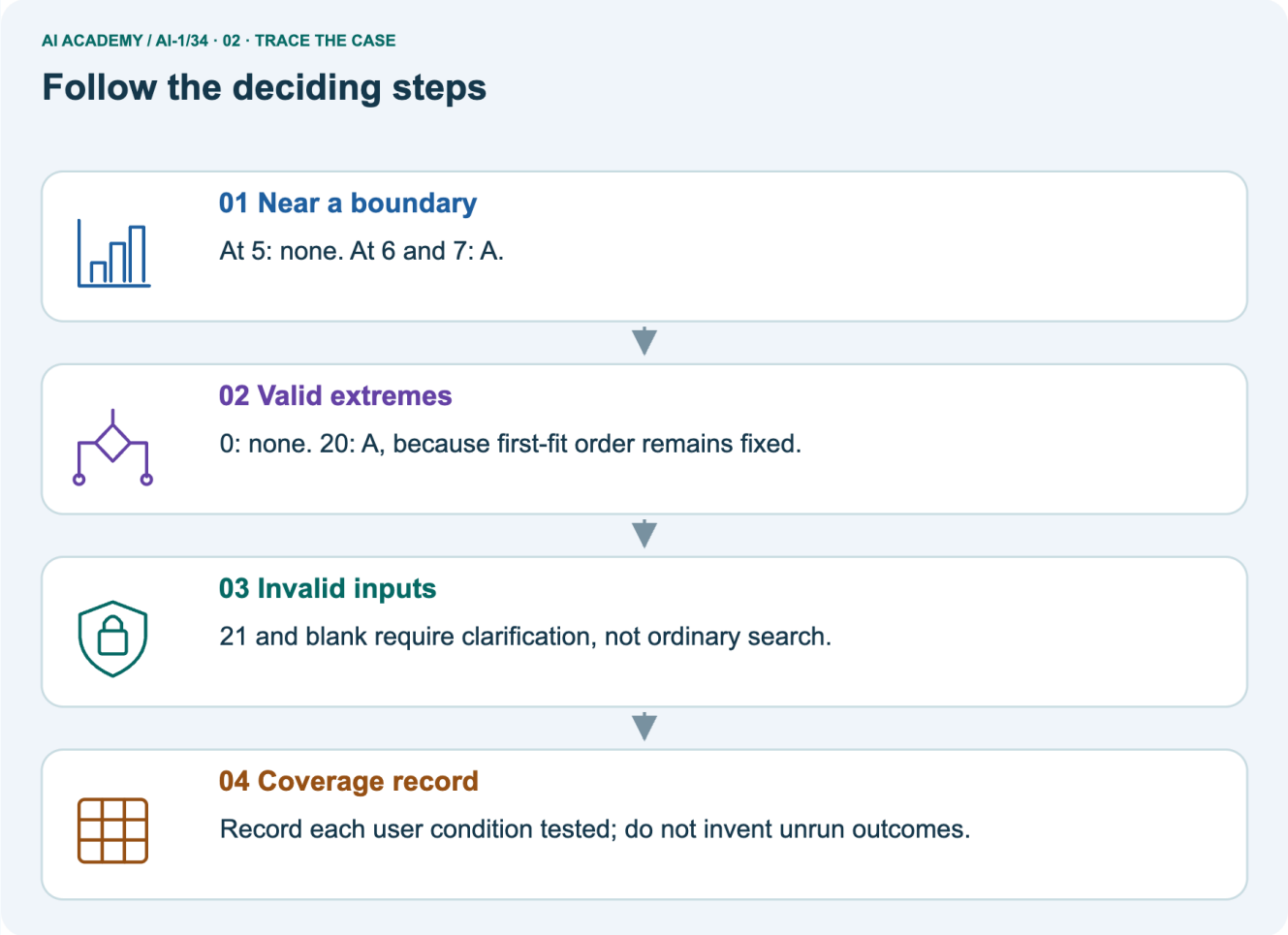
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional craft-book chooser accepts requests for whole available minutes from 0 to 20. Supplied facts: Booklet A takes 6 minutes; B takes 12. Return first duration at most available time.

Check 5,6,7; also 0,20,21 and blank. Same catalogue applies to every user.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



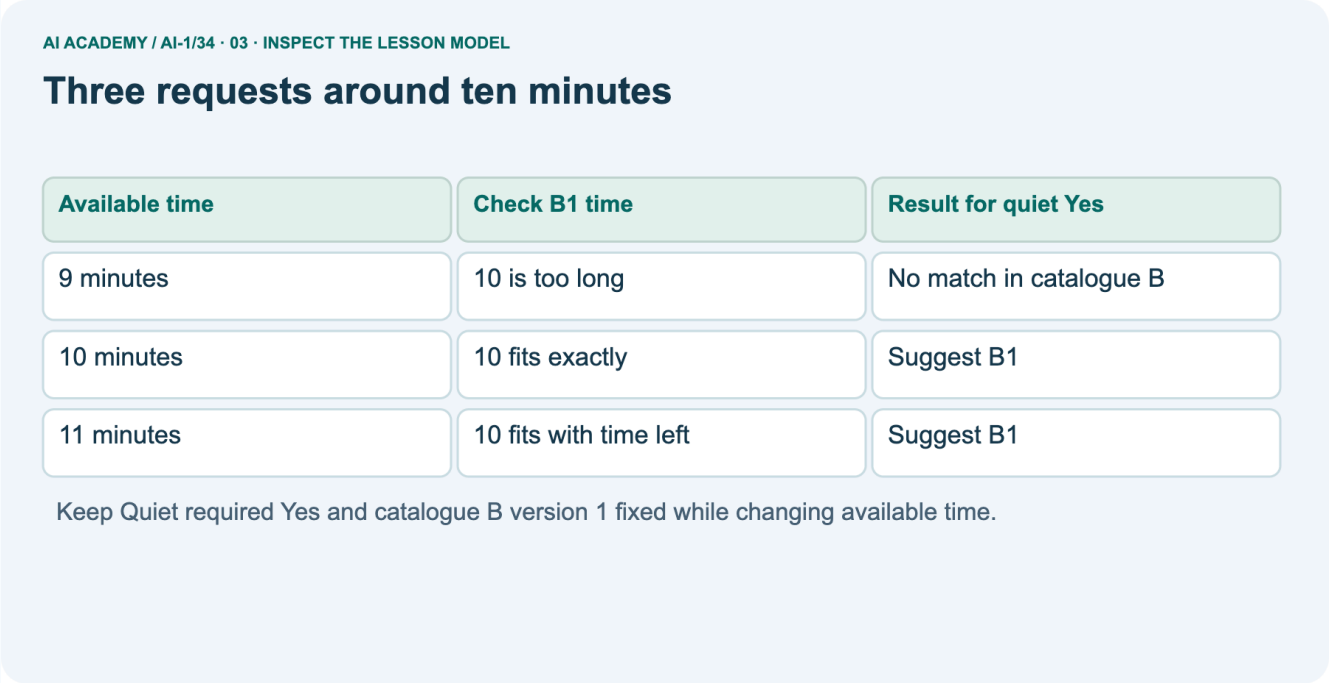
Follow the labelled picture with the matching explanation. Open the vector diagram

Near a **boundary**: At 5: none.

At 6 and 7: A. → Valid extremes: 0: none. 20: A, because first-fit order remains fixed. → **Invalid input**s: 21 and blank require clarification, not ordinary search. → Coverage record: Record each user condition tested; do not invent unrun outcomes.

Point to each numbered stage and explain which recorded **input** it uses.

03. Three requests around ten minutes

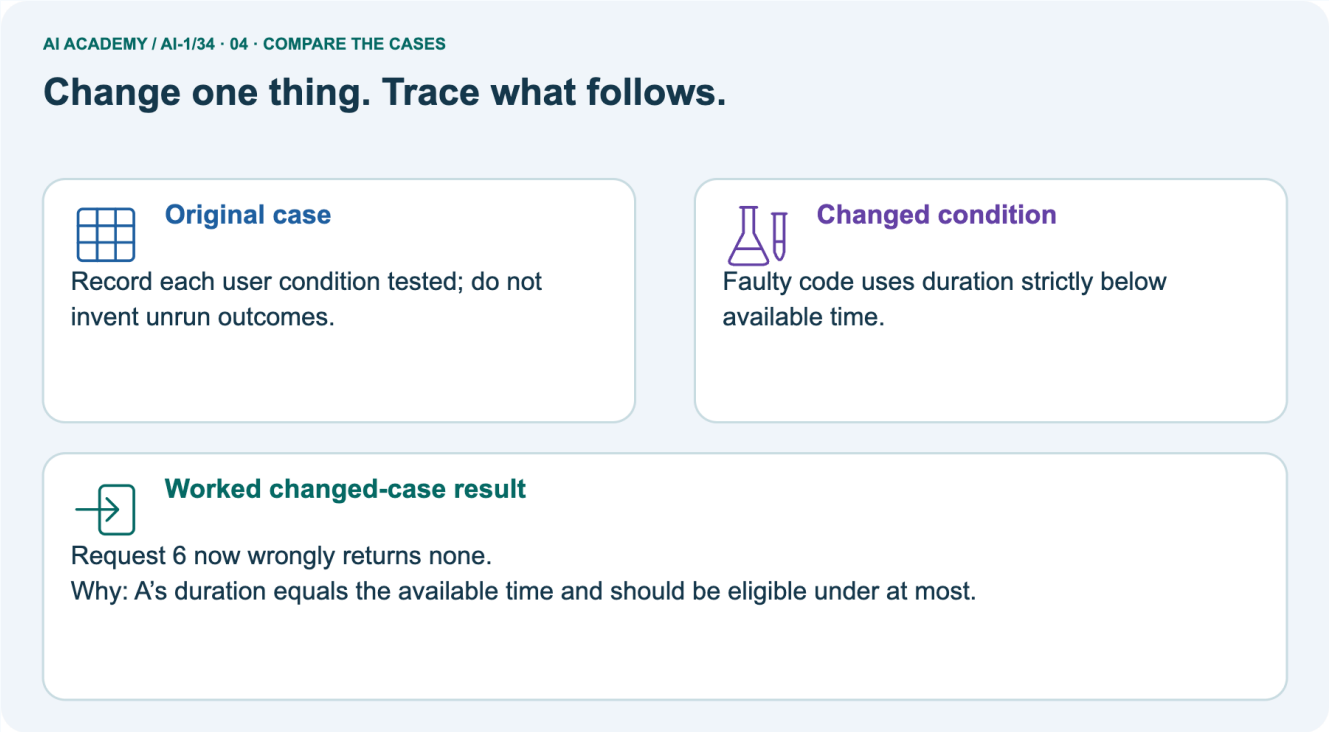


Follow the labelled picture with the matching explanation. Open the vector diagram

Three requests around ten minutes. 9 minutes | 10 is too long | No match in catalogue B. 10 minutes | 10 fits exactly | Suggest B1. 11 minutes | 10 fits with time left | Suggest B1

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



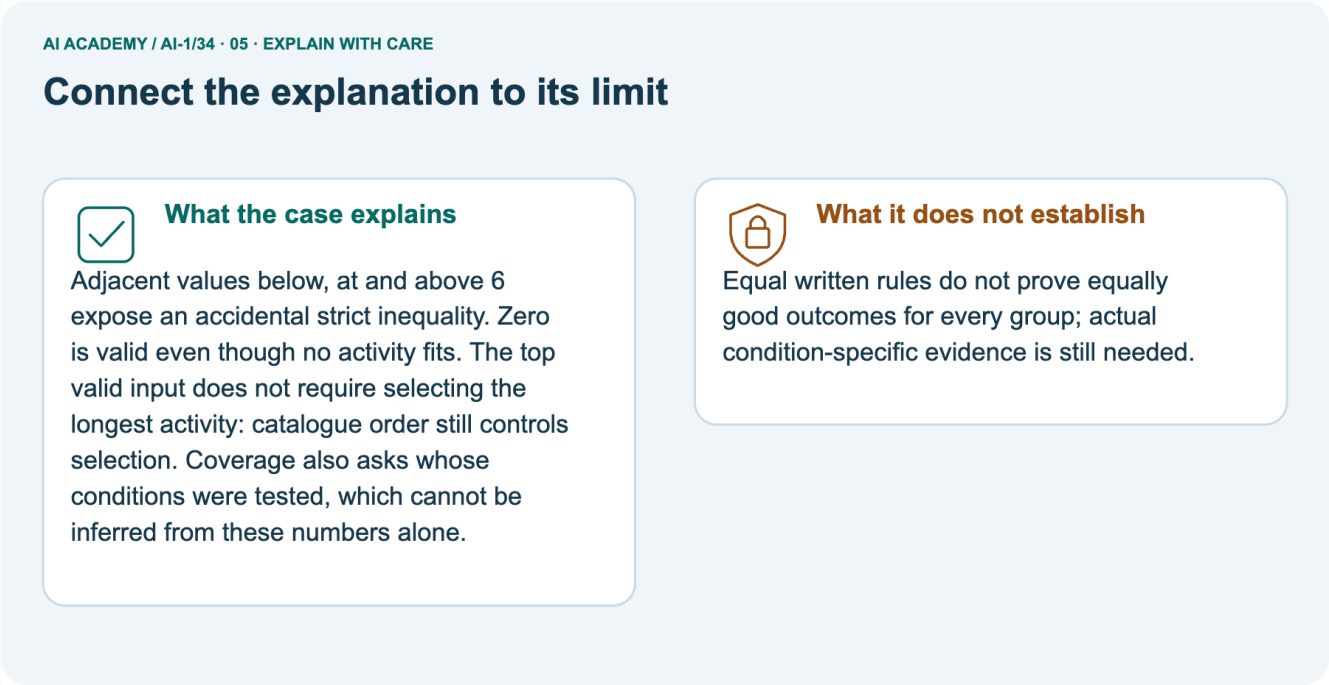
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Record each user condition tested; do not invent unrun outcomes. Changed condition: Faulty code uses duration strictly below available time. Expected result: Request 6 now wrongly returns none.

Reason: A's duration equals the available time and should be eligible under at most.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

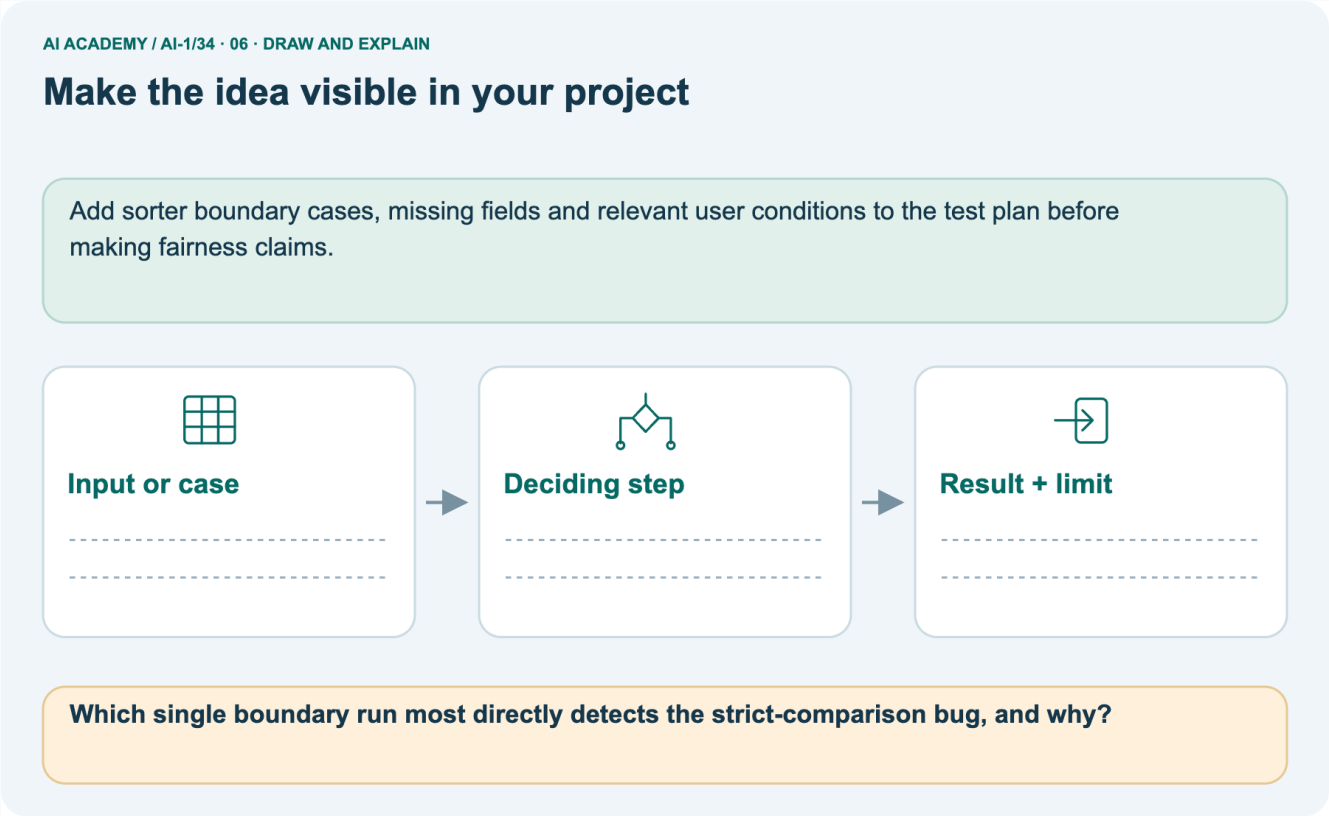
Adjacent values below, at and above 6 expose an accidental strict inequality. Zero is valid even though no activity fits.

The top valid **input** does not require selecting the longest activity: catalogue order still controls selection. Coverage also asks whose conditions were tested, which cannot be inferred from these numbers alone.

Limit: Equal **written rule**s do not prove equally good outcomes for every group; actual condition-specific **evidence** is still needed.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Add sorter **boundary** cases, missing fields and relevant user conditions to the test plan before making fairness claims. Draw your **input**, deciding step, result and limit.

Supported question: Which single **boundary** run most directly detects the strict-comparison bug, and why?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.